11 **(a)** e.g. noise can be eliminated / filtered / signal can be regenerated
extra bits can be added to check for errors
multiplexing possible
digital circuits are more reliable / cheaper
data can be encrypted for security
any sensible advantages, 1 each, max. 3 B3 [3]

(b) **(i)** **1.** higher frequencies can be reproduced B1 [1]

2. smaller changes in loudness / amplitude can be detected B1 [1]

(ii) bit rate $= 44.1 \times 10^3 \times 16$ C1
$= 7.06 \times 10^5\ s^{-1}$
number $= 7.06 \times 10^6 \times 340$
$= 2.4 \times 10^8$ A1 [2]

12 **(a)** **(i)** signal in one wire (pair) is picked up by a neighbouring wire (pair) B1 [1]